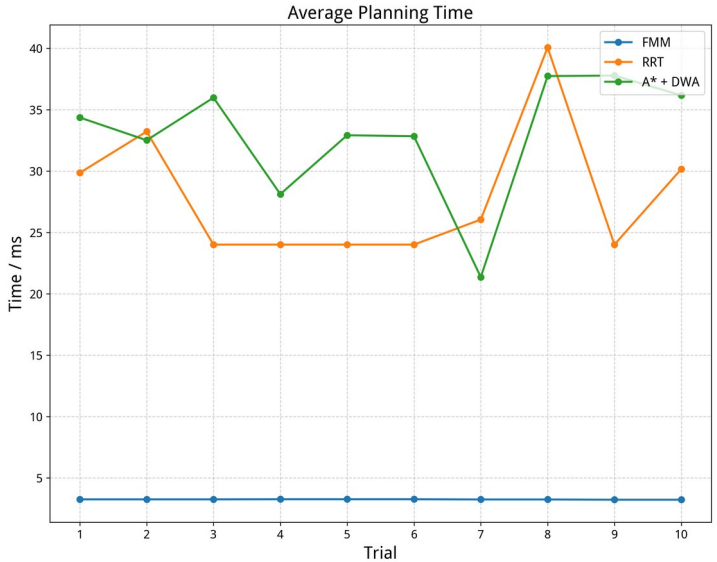
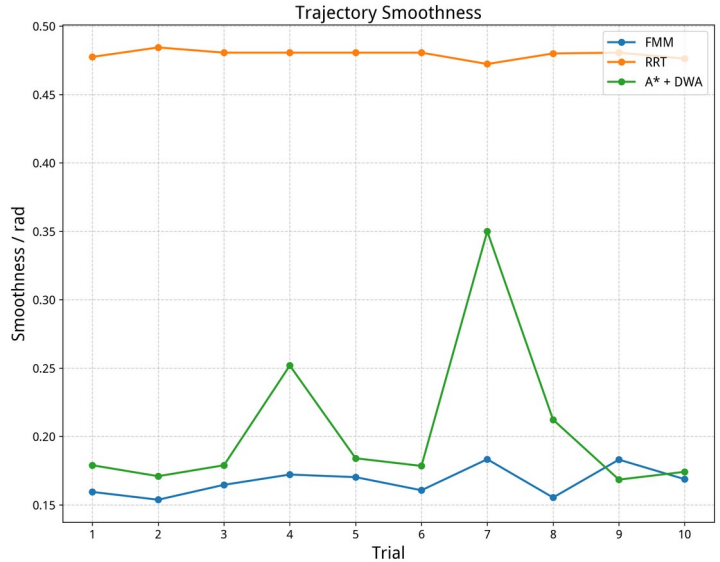
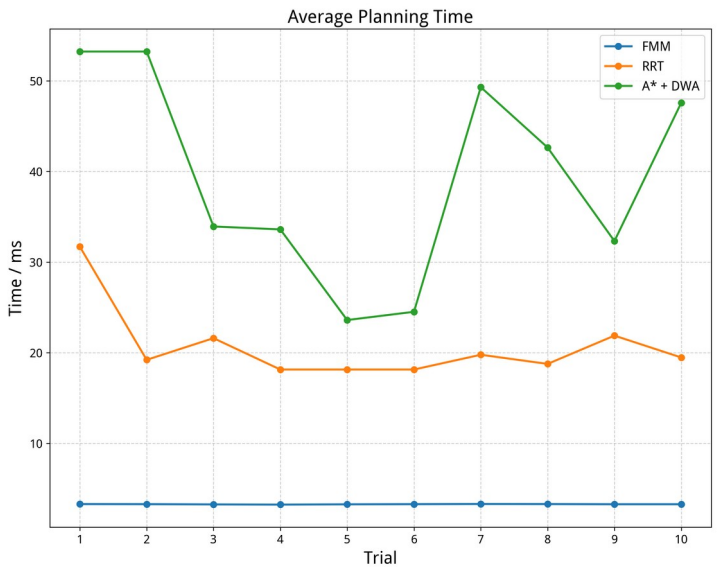

Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

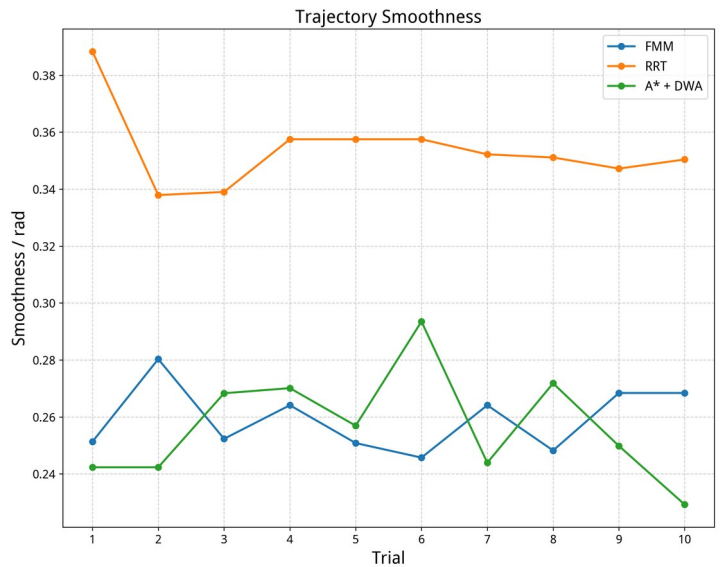
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

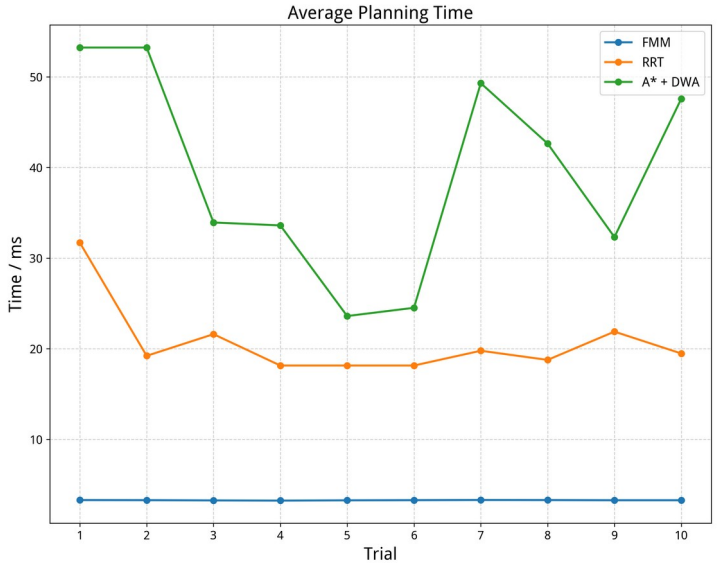
(b)


Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

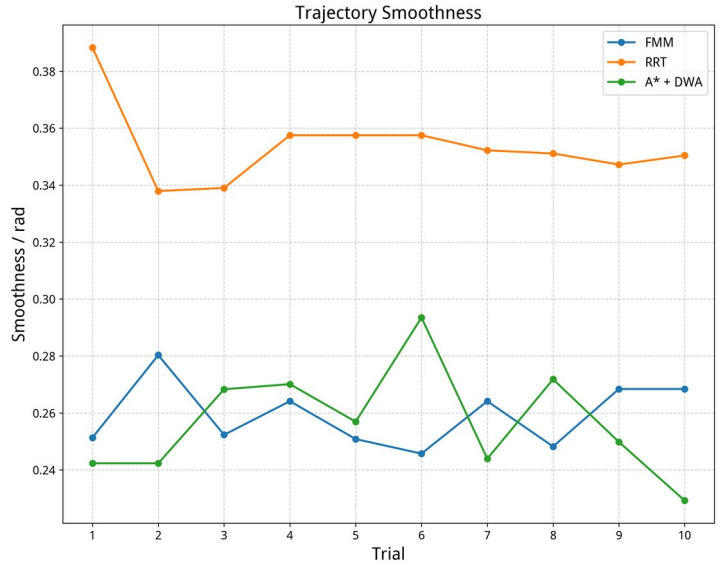
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

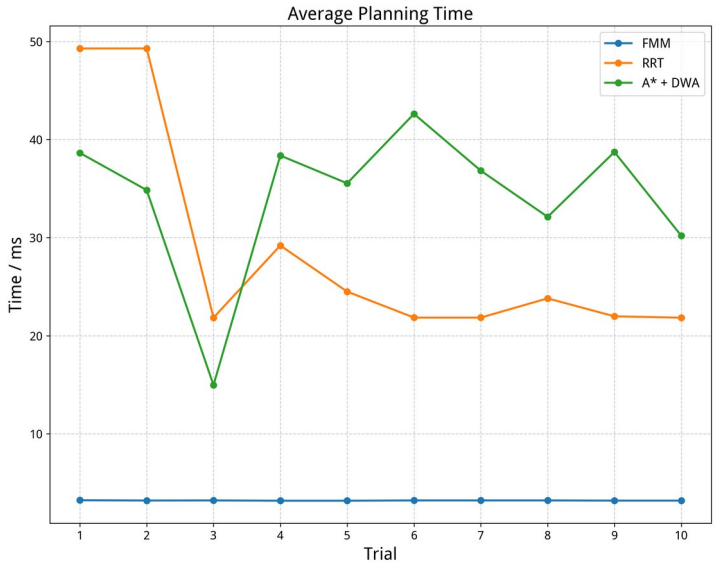
(b)


Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

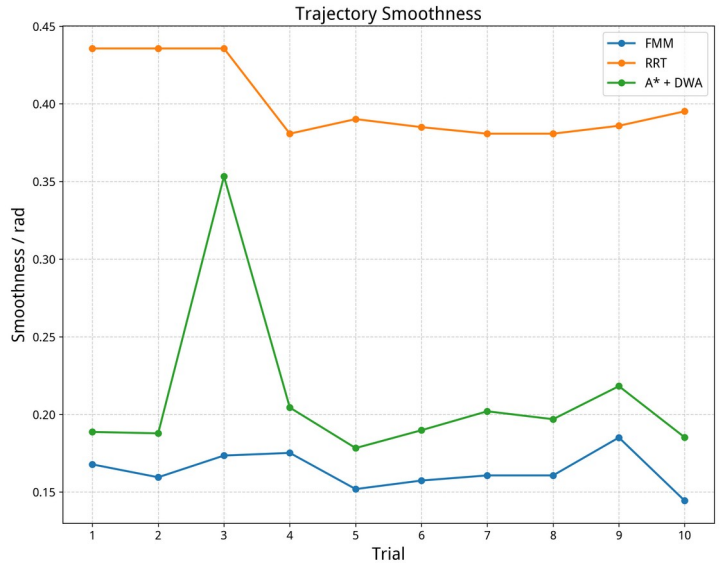
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

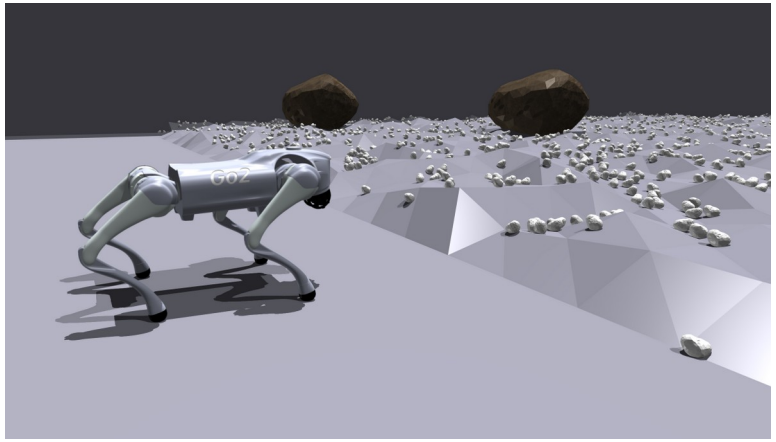
(b)


Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

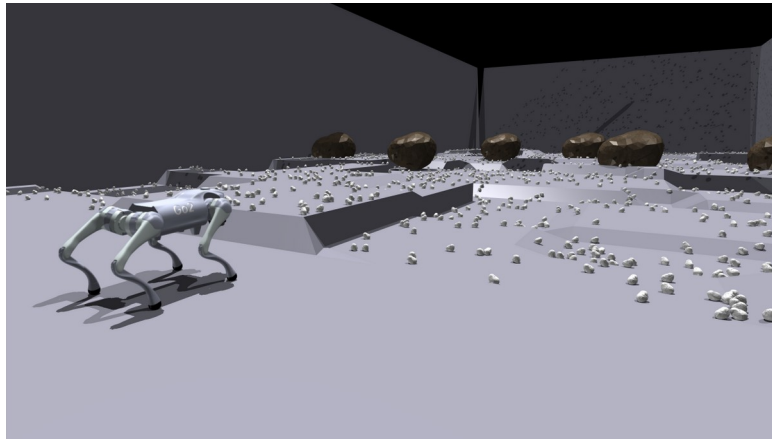
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

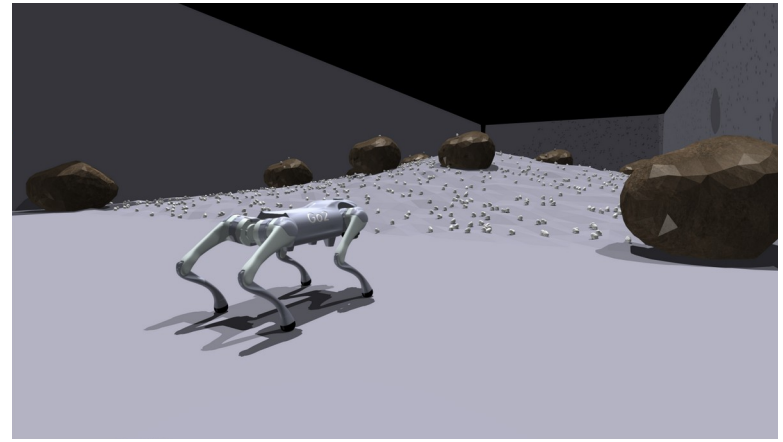
(b)



(a) Rough Slopes



(b) Discrete Obstacles



(c) Rough Flat





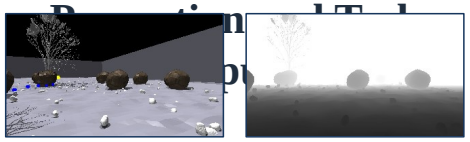
(a) Depth Image



(b) BEV Image

Hierarchical Visual Navigation and Locomotion Framework

Completed Chapter 3 system and evaluation pipeline



VIO Pose
(ideal or noisy)

Local Goal

Local Perception and Planning

RGB-D Point-cloud Projection → Local BEV Height and Obstacles → Dual-layer Cost Map Hard Collision + Soft Cost

FMM Reference Corridor → Footprint MPPI Trajectory Optimization → Velocity Command (vx, wz)

Locomotion Inputs

Desired Base Velocity

Proprioceptive State

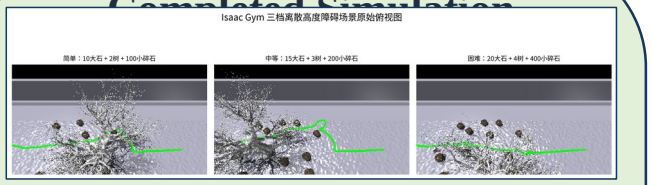
Historical Observations

HIMLoco Locomotion Controller

State and History Encoder → **HIMLoco Policy Network** → Joint Position Targets

Shared locomotion policy for all navigation planners

Completed Simulation



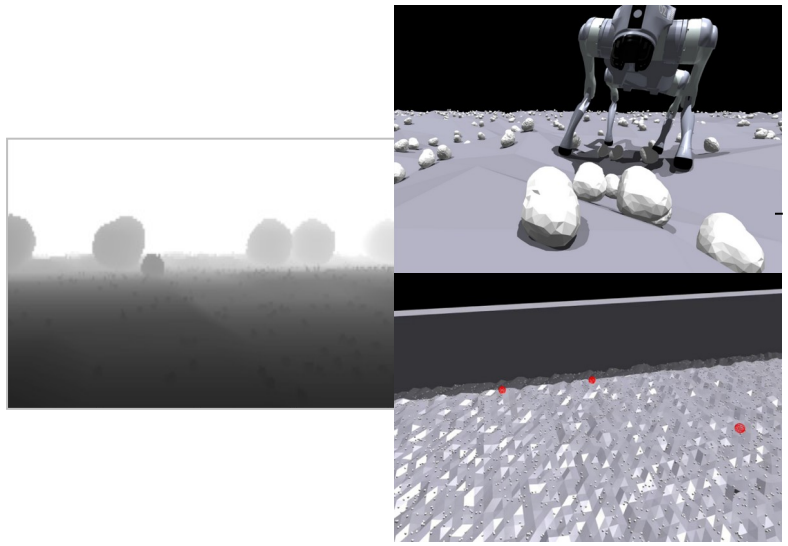
Three difficulty levels
30 fixed seeds per method

Internal ablation
Fixed FMM, Dual-layer FMM, Footprint MPPI

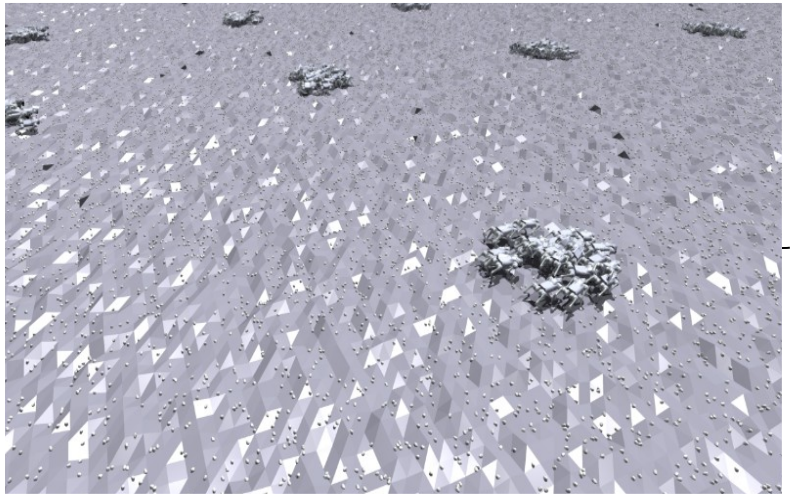
External comparison
iPlanner, PS-DWA, SAPF

VIO robustness
ideal, mild, medium, severe

750 formal closed-loop trials



Depth Camera Perception Driving



Reward Function Driving

Trajectory Planning

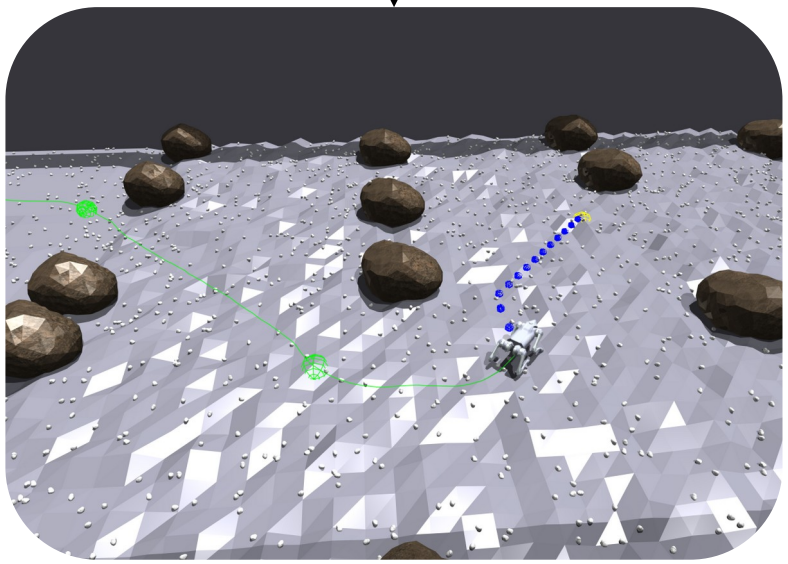
Reinforcement Learning

Navigation Controller

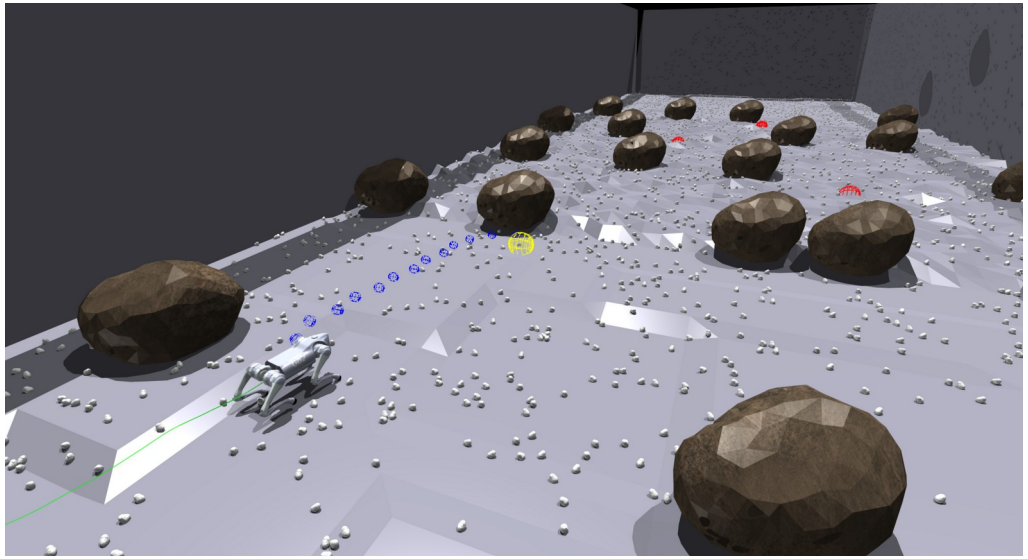
v_x, ω_{yaw}

Gait Controller

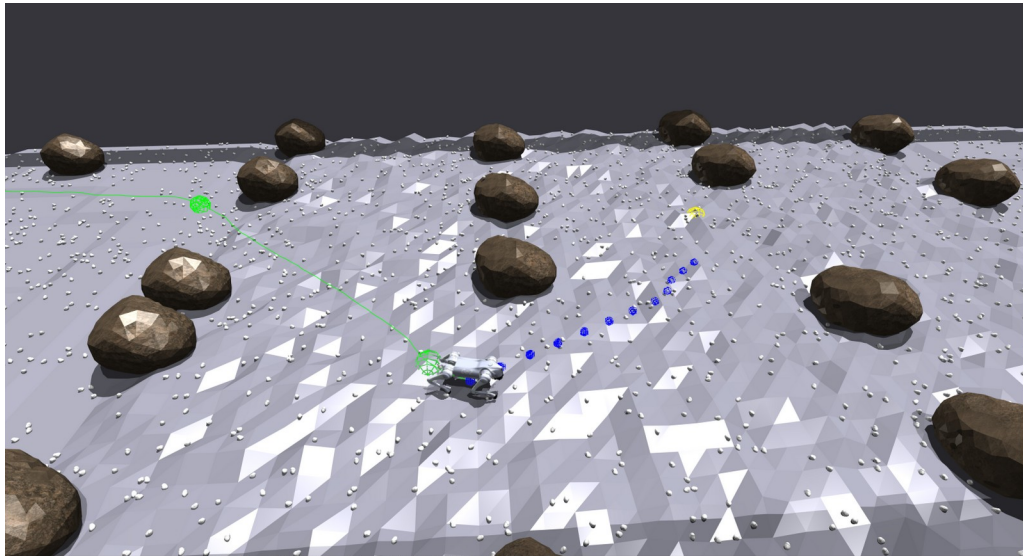
Joint Position Commands



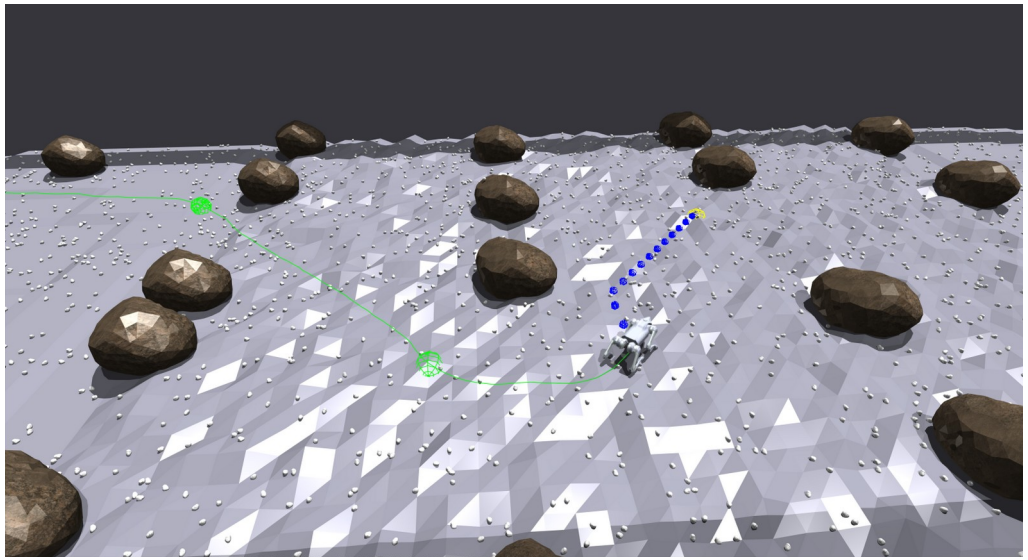
Autonomous Navigation



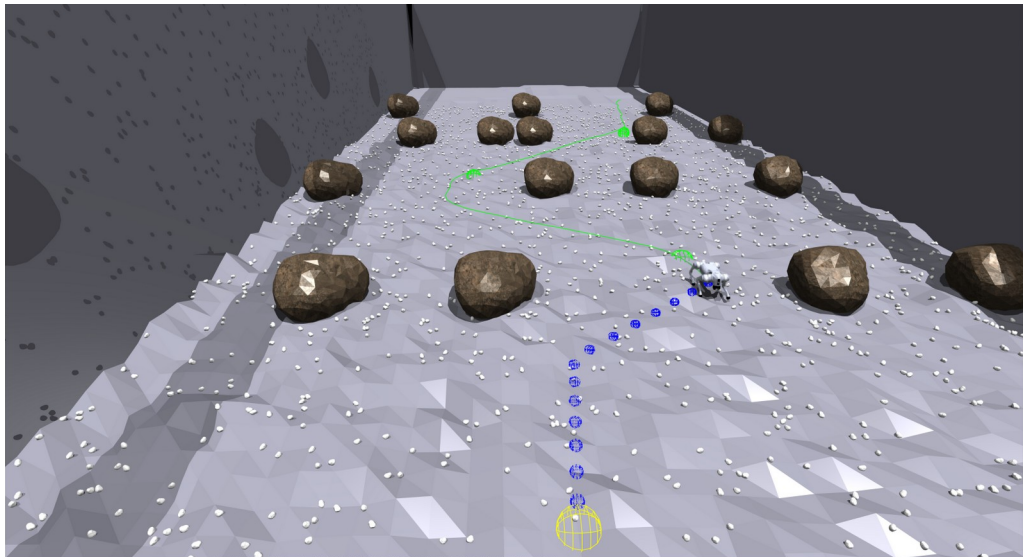
(a)Start Point — Waypoint A



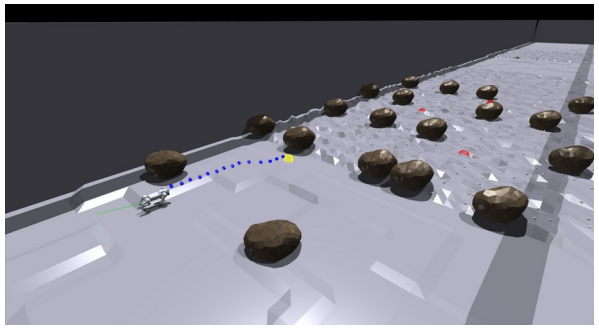
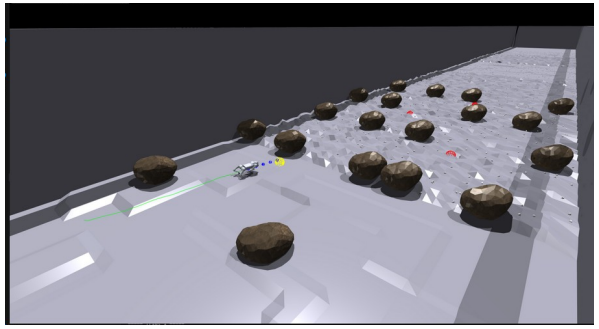
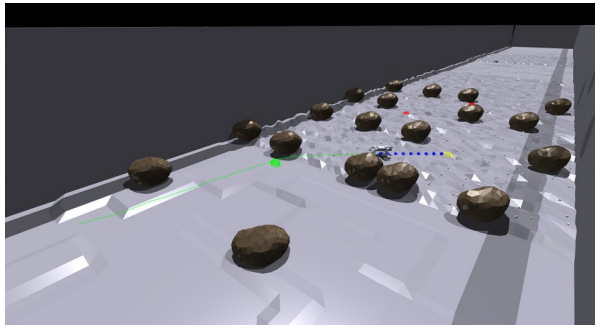
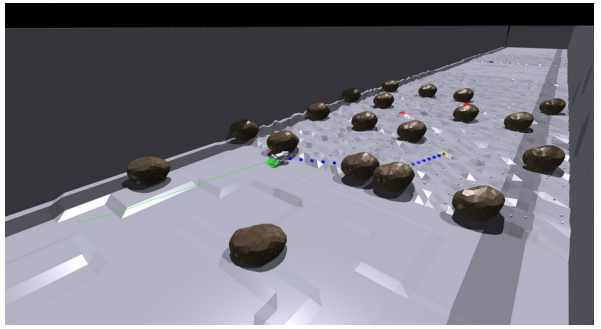
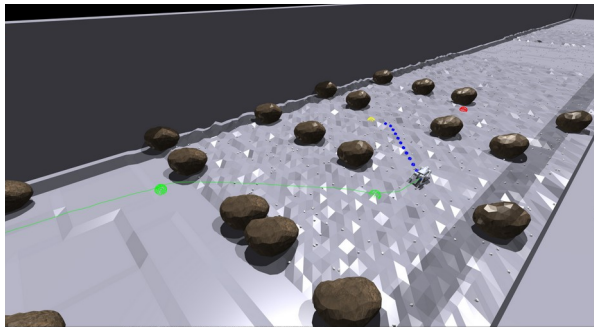
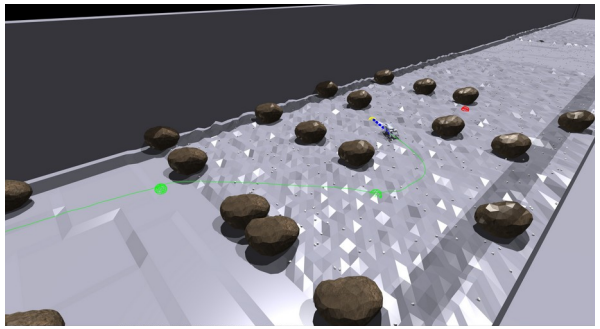
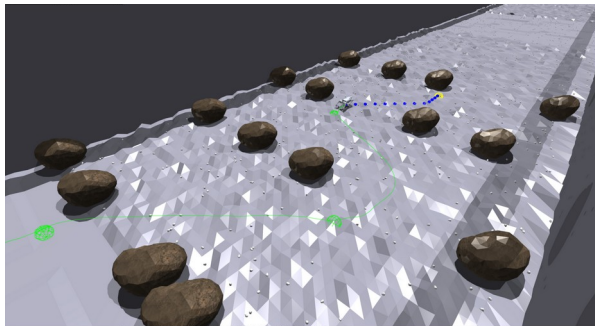
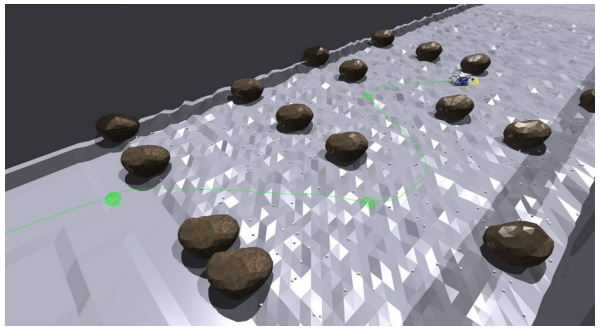
(b) Waypoint A — Waypoint B

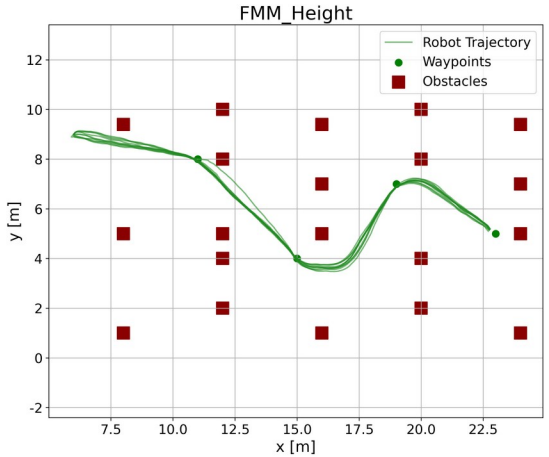


(c)Waypoint B — Waypoint C

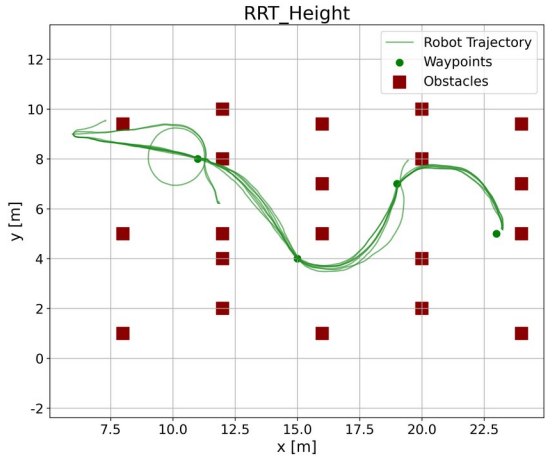


(d)Waypoint C — End Point

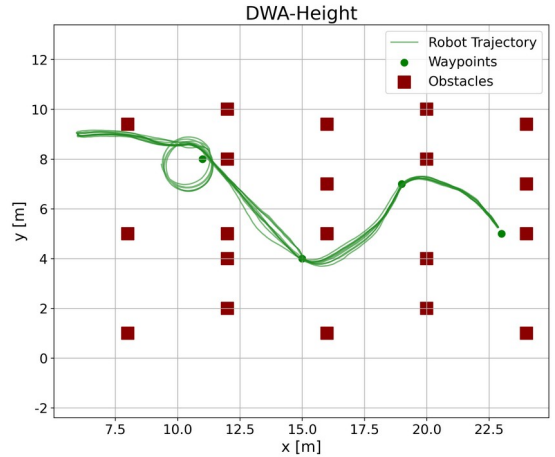










FMM_Height
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

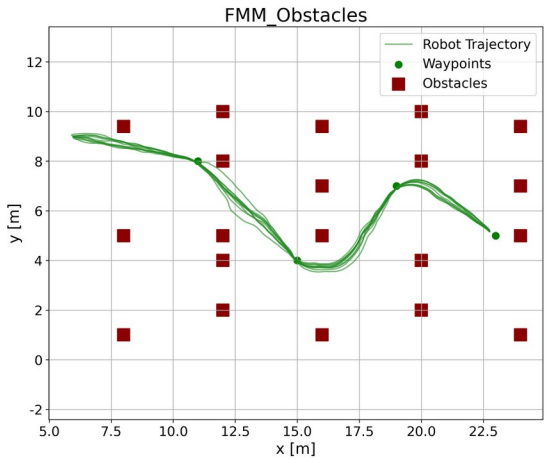
(a)


RRT_Height
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

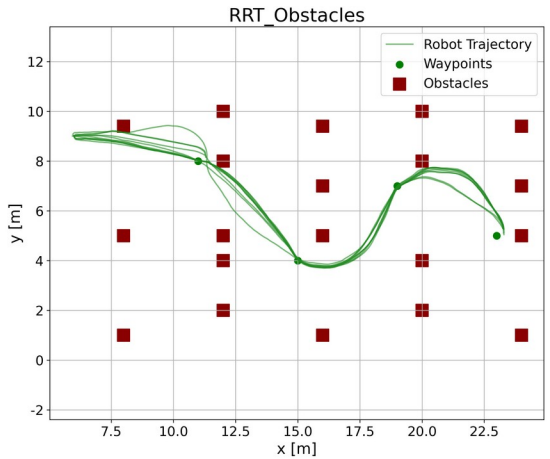
(b)


DWA-Height
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

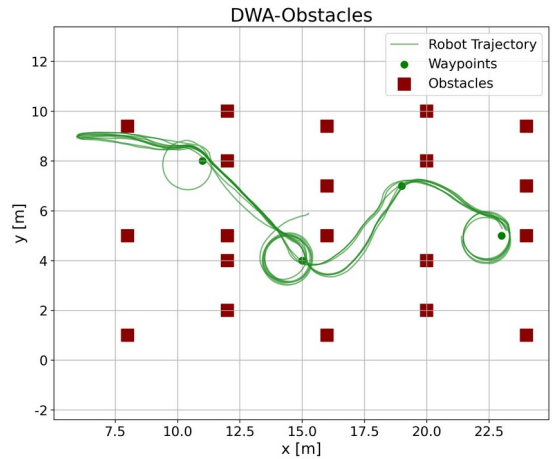
(c)


FMM_Obstacles
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

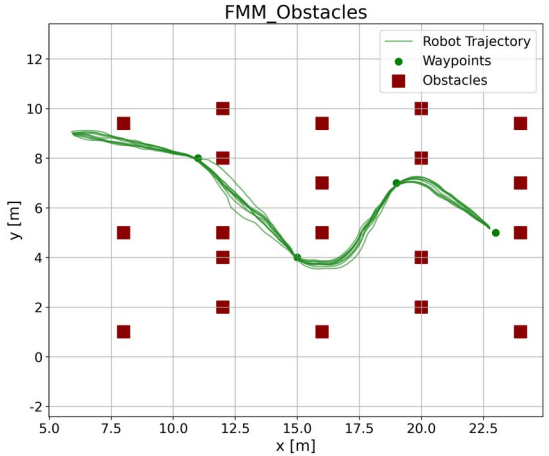
(a)


RRT_Obstacles
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

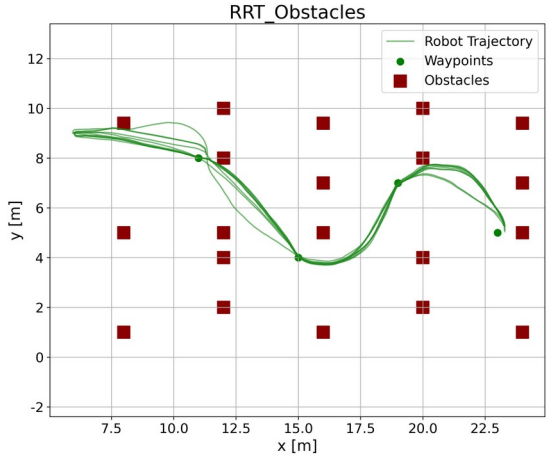
(b)


DWA-Obstacles
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

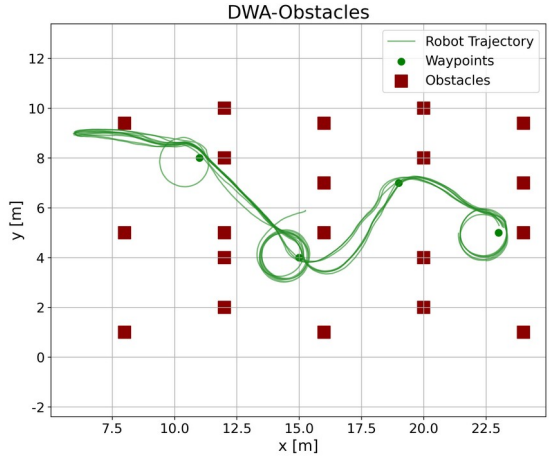
(c)


FMM_Obstacles
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

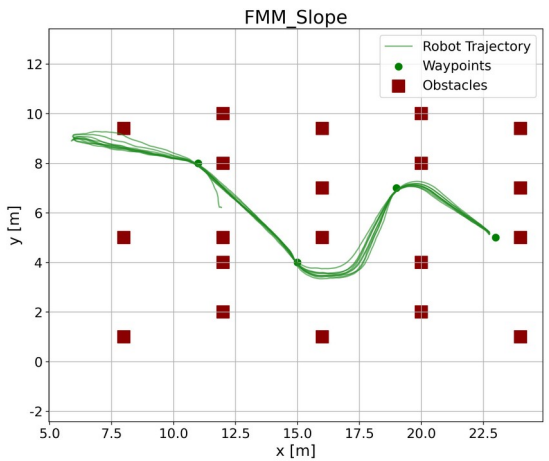
(a)


RRT_Obstacles
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

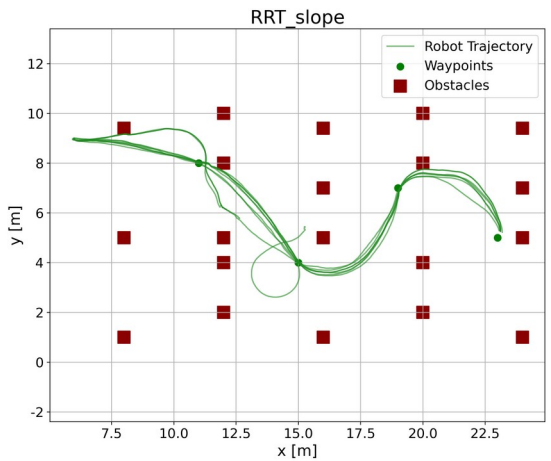
(b)


DWA-Obstacles
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

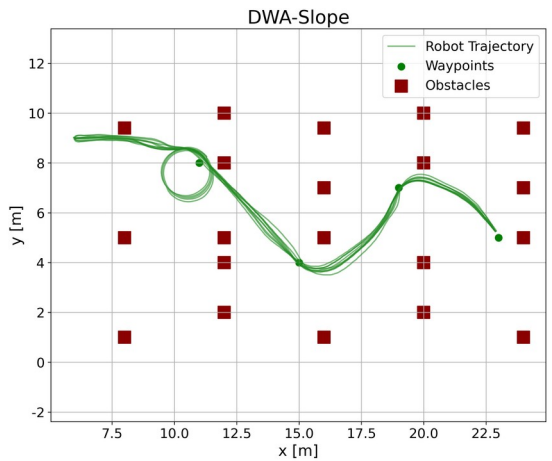
(c)


FMM_Slope
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

(a)


RRT_slope
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

(b)


DWA-Slope
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

(c)